GRPO unraveled: A Small-Scale Empirical Study of RLVR
CMPT 981 Project Proposal

Ankush Singh
Simon Fraser University
asa715@sfu.ca

Simon Doan
Simon Fraser University
vtd@sfu.ca

Azaan Rehman
Simon Fraser University
ara199@sfu.ca

Abstract

Reinforcement learning with verifiable rewards (RLVR), and in particular Group Relative Policy Optimization (GRPO), is now the standard recipe for training reasoning in large language models. Recent theory makes concrete predictions about its dynamics: how many sampled groups carry a learning signal, whether GRPO amplifies the success probability, and when a step size is too large. These predictions are derived for exact gradients and idealized policies, and are hard to check at LLM scale, where every run is expensive and the true success probability is unknown. We propose an experiment-centred project that tests them on a small synthetic task with a binary verifier. Its difficulty is set exactly by the answer length, and the policy's success probability can be computed in closed form, so runs take seconds on a CPU. We vary the initial success rate, the number of sampled answers per prompt, reward sparsity, KL strength, and step size, and compare each result with the corresponding theoretical prediction. As a small theoretical contribution, we extend the success-amplification fixed point of Mroueh [2025] from infinite to finite group size, which gives closed forms for mean-only and leave-one-out baselines and a testable prediction of how the group size shifts GRPO's fixed point. We also show that on-policy SGD shares these fixed points with Mroueh's exact iteration but not their stability. We close with a confirmation on a 0.5B-parameter LLM.

1 Introduction and Motivation

Reinforcement learning (RL) learns a policy of actions from a reward signal that is often sparse and delayed. In the context of large language models (LLMs), RL is used to align the model's responses with human preferences or task-specific objectives, most notably through reinforcement learning from human feedback (RLHF) [Ouyang et al., 2022]. However, RLHF can be expensive and time-consuming, as it requires humans to label which responses they prefer, and those labels are used to train a reward model.

In 2025, DeepSeek-R1-Zero [Guo et al., 2025] showed that a language model could learn long chain-of-thought reasoning from a rule-based reward, mainly a binary check of whether the final answer is correct, without any human feedback. This approach is called reinforcement learning with verifiable rewards (RLVR). The model was trained using Group Relative Policy Optimization (GRPO) [Shao et al., 2024], which samples multiple answers per prompt and normalizes the advantages by the group standard deviation. Since then, practitioners have relied on rules of thumb, such as sampling more answers for problems with a low success rate, normalizing advantages by the group metrics, and averaging the loss over tokens. However, each of these choices changes the fundamental optimization problem, and it is still unclear which ones matter and why.

Initial theoretical work provides precise answers in simplified settings [Mroueh, 2025, Suk and Duan, 2025, Thrampoulidis et al., 2026, Bay and Yearick, 2026]. Examples include the success amplification property of GRPO [Mroueh, 2025], which states that, with a KL penalty, the success probability settles at a fixed point higher than that of the reference model, and the gradient gap [Suk and Duan, 2025], the difference between the expected score of correct and incorrect responses, which is used to derive a step-size threshold below which training converges. However, these results are derived under simplified conditions, far from the stochastic, finite-sample regime in which GRPO is actually run. For example, they assume exact expectations and, in the case of Mroueh [2025], an exactly solved inner optimization problem. Testing them directly is also hard, because LLM experiments cannot isolate a single variable: the success rate, task difficulty, and response length all change together.

Our project aims to bridge the gap between theory and practice. We plan controlled experiments that will allow us to sweep a grid of parameters, including the initial success rate, group size, reward sparsity, KL strength, and step size. By changing one parameter at a time and observing the effect on the learning dynamics, we can test the theoretical predictions in a setting closer to practice, with finite samples, stochastic gradients, and a learned policy. These results will offer further insight into the dynamics of RLVR and GRPO in practical settings. This can lead to a better understanding of the optimization problem and help guide how these settings are chosen when training LLMs with verifiable rewards.

2 Related Work

GRPO and RLVR. GRPO [Shao et al., 2024] drops the value network of Proximal Policy Optimization (PPO). For each prompt, it samples a group of G answers and uses the group-normalized reward $(r_i - \bar{r})/\mathrm{std}(r)$ as the advantage. DeepSeek-R1 [Guo et al., 2025] scaled this up with rule-based verifiers, $G = 16$, and a small KL coefficient that keeps the policy close to a reference model. A later study, Dr. GRPO [Liu et al., 2025], argued that the std-normalization and the per-response length normalization introduce biases. RLOO [Ahmadian et al., 2024] instead uses a leave-one-out baseline, the mean reward of the other $G - 1$ answers, which reduces variance without biasing the gradient. We use all three as advantage variants in our experiments to see how they affect the learning dynamics.

Success amplification [Mroueh, 2025]. In 2025, Mroueh analyzed GRPO in the RLVR setting and identified its success amplification property. The paper shows that, for binary rewards with a reverse-KL penalty and no clipping, GRPO's objective can be seen as a contrastive loss between correct and incorrect samples taken from the previous policy. Solving the KL-regularized problem exactly at each iteration shows that the success probability follows a recurrence $p_n = h(p_{n-1})$ for an explicit function h. Every fixed point of this recurrence satisfies $p^* > p_{\mathrm{ref}}$, which means that GRPO increases the success probability above that of the reference model. In this project, we will test whether SGD training reaches the predicted p^* for different values of β (the strength of the KL penalty) and for finite G. We will also compare this reverse-KL penalty, $\mathrm{KL}(\pi_\theta \,\|\, \pi_{\mathrm{ref}})$, with the forward-KL penalty $\mathrm{KL}(\pi_{\mathrm{ref}} \,\|\, \pi_\theta)$, which will cover the difference between mode-seeking and mass-covering policies.

Gradient gap and step-size thresholds [Suk and Duan, 2025]. The paper defines the gradient gap, the difference between the expected score of correct and incorrect responses, and proves a sufficient step-size condition for convergence together with lower-bound constructions where a constant step makes performance decrease to zero. With exact gradients, the threshold scales as $1/T$ in the response length T and shrinks with the success rate through $(1 - J)$; the paper argues that GRPO's $1/|y|$ normalization provides exactly this scaling and Dr. GRPO's does not. Its experiments use a single step size. We test whether a sharp transition appears in a stochastic, learned-policy setting, and how it moves with response length when responses of different lengths share one learning rate. Ge et al. [2026] sharpen this question: they show that std-normalization acts as an adaptive step size matched to local curvature, with a speed-up that depends on how much the per-prompt reward variance differs across prompts.

Group size and informative prompts. A group with all-equal rewards has zero advantage, and a common practice is to filter out such groups. Bay and Yearick [2026] show that GRPO, Dr. GRPO and DAPO differ only in how they use the group standard deviation, derive a closed-form group-size law in $G\,p(1-p)$, and check it on a Bernoulli toy model at $G = 8$ and by subsampling logged

rollouts. Nie et al. [2026] prove that when the success rate varies across prompts, the fraction of all-equal groups exceeds the prediction from the average success rate, and that a fixed-reference advantage $A = 2r - 1$ outperforms Dr. GRPO at $G = 4$ on GSM8K. Plyusov et al. [2026] show that a group can have mixed rewards yet miss rare correct solutions, with a non-monotonic dependence on G.

We do not claim the informative-group fraction $1 - p^G - (1-p)^G$ or the $p(1-p)$ scaling as new. The closest prior work to ours is Bay and Yearick [2026], Nie et al. [2026] and Plyusov et al. [2026], each of which varies G in theory or in a toy model. Our contribution is a controlled *training* grid over G, p_0, KL strength and step size, with every quantity measured exactly, compared against these closed forms.

Pass@K advantage shaping. Thrampoulidis et al. [2026] show that advantage-shaping heuristics, such as hard-example upweighting, implicitly maximize surrogate rewards such as pass@K, and derive an unbiased leave-one-out pass@K advantage. We test whether optimizing pass@K preserves response diversity and at what cost to pass@1.

Entropy and diversity collapse. Yue et al. [2025] show that RLVR-trained models beat their base model at small k but lose to it at large k, which suggests that RLVR mostly sharpens what the base model can already do.

3 Problem Formulation

We start with a prompt x and a policy $\pi_\theta(y \mid x)$ that generates a response y. A verifier returns a reward $r(x, y) \in \{0, 1\}$, and the policy succeeds on prompt x with probability $p(x)$. We sample a group of G responses per prompt, compute advantages A_i within the group, and take a gradient step to increase

$$J(\theta) = \mathbb{E}[r] - \beta \, \mathrm{KL}(\pi_\theta \,\|\, \pi_{\mathrm{ref}}). \tag{1}$$

We vary the following variables: the initial success rate p_0; the group size G; the reward sparsity, set by the answer length L and vocabulary size K (a uniform policy succeeds with probability K^{-L}, and a short supervised warm-start lets us set p_0 independently); the KL strength β; the step size η; and the advantage type (GRPO, mean-only, RLOO, or pass@k).

We use three policies of increasing realism. *Bandit:* $L = 1$, with a separate softmax over the K options for each prompt, so the theory's assumptions hold except for finite G and stochastic gradients. *Synthetic task:* $L \geq 1$, and the policy is a product of L softmaxes computed by one network shared across prompts; a multi-solution variant, with several correct tokens per position, lets us measure diversity. *Small LLM:* Qwen2.5-0.5B-Instruct on GSM8K [Cobbe et al., 2021], a real-world example in which $p(x)$ is unknown and must be estimated by sampling.

4 Plan

Our main goal is to test the limits of the theoretical predictions described in the related work. Those works assume simplified conditions; we will see where their predictions break when those conditions are not fulfilled.

We will run a grid of experiments over the variables defined in Section 3. Ideally we will cover the entire grid, but we prioritize finding where the theoretical predictions break, and then sample the grid more densely around those limits.

We define the following core experiments, in order:

E1 Initial success rate $\times$ group size. Our first test checks the effects of the initial success rate and the group size on the learning speed. We will also compare our measured learning speeds with the group-size law of Bay and Yearick [2026].

E2 Success amplification [Mroueh, 2025]. In the simpler bandit setting ($L = 1$), we will test the success amplification property of GRPO by comparing the fixed points measured at finite group size with the infinite-group fixed points of Mroueh [2025], for both the reverse- and the forward-KL penalty.

E3 Step-size threshold [Suk and Duan, 2025]. We will test the step-size threshold on the synthetic task. In the bandit, a GRPO step cannot lower the success rate at any step size, so a threshold can only appear with a shared network. We will sweep the learning rate over a log grid and look for a narrow range where success first rises and then stalls or collapses. We will also check whether the threshold shrinks roughly as $1/L$ and as p_0 grows.

E4 Pass@k and response diversity [Thrampoulidis et al., 2026]. On the multi-solution task, we will compare pass@k advantages with GRPO in terms of pass@1, pass@k, and response diversity.

E5 Escaping very sparse rewards. When rewards are so sparse that almost no group has mixed rewards, training stalls. We will test what helps the model escape this regime.

E6 Small-LLM check. We will train a small LLM to check whether the theoretical predictions and the insights from the previous experiments carry over to a real model.

Current status. We have derived the finite-group fixed point (Section A) and implemented a one-prompt GRPO bandit whose runs settle near the predicted fixed points. The full training harness (synthetic task, shared network, advantage estimators, and exact metrics) is not yet implemented; it is the first milestone.

Timeline and roles. Weeks 1–2: build the harness and unit-test the exact metrics against the closed-form predictions; complete E1. Weeks 3–4 (milestone): complete E2 and E3, and set up the E6 pilot run. Weeks 5–6: E4–E6, ablations, and the final report. Each member owns one experiment end-to-end (implementation, runs, figures, and the related section of the report), and all three share the code and the literature review.

What we expect to learn. The theory is derived for exact gradients and simple policies. Where our curves deviate from it, we will identify the cause: finite G (which our finite-group fixed point quantifies for E2), stochastic batches, function approximation shared across prompts, or the factorized multi-token policy. These deviations are the most useful output of the project.

References

Arash Ahmadian, Chris Cremer, Matthias Gallé, et al. Back to basics: Revisiting REINFORCE-style optimization for learning from human feedback in LLMs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.

Yong Yi Bay and Kathleen A. Yearick. GRPO, Dr. GRPO, and DAPO are three operations on one number: The group-standard-deviation identity. *arXiv preprint arXiv:2607.00152*, 2026.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Cheng Ge, Caitlyn Heqi Yin, Hao Liang, and Jiawei Zhang. Why GRPO needs normalization: A local-curvature perspective on adaptive gradients. *arXiv preprint arXiv:2601.23135*, 2026.

Daya Guo, Dejian Yang, Haowei Zhang, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *Nature*, 645:633–638, 2025. arXiv:2501.12948.

Zichen Liu, Changyu Chen, Wenjun Li, et al. Understanding R1-Zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025.

Youssef Mroueh. Reinforcement learning with verifiable rewards: GRPO's effective loss, dynamics, and success amplification. *arXiv preprint arXiv:2503.06639*, 2025.

Wenhua Nie, Jianan Wu, Junlin Liu, et al. Gradient starvation in binary-reward GRPO: Why group-mean centering fails and why the simplest fix works. *arXiv preprint arXiv:2605.07689*, 2026.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, 2022. arXiv:2203.02155.

Daniil Plyusov, Alexey Gorbatovski, Boris Shaposhnikov, Viacheslav Sinii, Alexey Malakhov, Daria Korotyshova, and Daniil Gavrilov. F-GRPO: Don't let your policy learn the obvious and forget the rare. *arXiv preprint arXiv:2602.06717*, 2026.

Zhihong Shao, Peiyi Wang, Qihao Zhu, et al. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

Joe Suk and Yaqi Duan. On the optimization dynamics of RLVR: Gradient gap and step size thresholds. *arXiv preprint arXiv:2510.08539*, 2025.

Christos Thrampoulidis, Sadegh Mahdavi, and Wenlong Deng. Advantage shaping as surrogate reward maximization: Unifying pass@K policy gradients. *Transactions on Machine Learning Research*, 2026. arXiv:2510.23049.

Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, et al. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? In *Advances in Neural Information Processing Systems*, 2025. arXiv:2504.13837.

A Proof of ??

Fix a prompt and write $\pi = \pi_\theta(\cdot \mid x)$ and $p = \sum_{y:r(y)=1} \pi(y)$. The advantages are computed from sampled rewards and are not differentiated, so the expected update direction of the reward term is $g = \mathbb{E}\big[\frac{1}{G}\sum_{i=1}^{G} A_i \nabla_\theta \log \pi(y_i)\big]$. The group is exchangeable, so $g = \mathbb{E}[A_1 \nabla_\theta \log \pi(y_1)]$. Conditional on y_1, the other responses are independent of y_1, so $\mathbb{E}[A_1 \mid y_1]$ depends on y_1 only through $r(y_1)$, and equals $w(1) = a_G^+(p)$ or $w(0) = -a_G^-(p)$. Hence

$$g = \sum_y \pi(y)\, w(r(y))\, \nabla_\theta \log \pi(y) = \nabla_\theta \sum_y \pi_\theta(y)\, w(r(y))\Big|_{w \text{ fixed}}.$$

Adding the exact KL penalty, a stationary point is a stationary point of $F(\pi) = \sum_y \pi(y) w(r(y)) - \beta\, \mathrm{KL}(\pi \,\|\, \pi_{\mathrm{ref}})$ at fixed w. When the policy class contains every distribution with full support (a softmax over all responses), $\nabla_\theta F = 0$ if and only if $w(r(y)) - \beta\big(\log \frac{\pi(y)}{\pi_{\mathrm{ref}}(y)} + 1\big)$ is constant in y, i.e. $\pi(y) \propto \pi_{\mathrm{ref}}(y) \exp(w(r(y))/\beta)$. Summing over correct and incorrect responses,

$$\frac{p}{1-p} = \frac{p_{\mathrm{ref}}}{1-p_{\mathrm{ref}}} \exp\Big(\frac{a_G^+(p) + a_G^-(p)}{\beta}\Big),$$

which is Eq. (**??**), with p evaluated at the stationary point itself.

For `mean-only`, $A_1 = r_1 - \frac{1}{G}\sum_i r_i$, so $a_G^+ = 1 - \frac{1+(G-1)p}{G} = \frac{G-1}{G}(1-p)$ and $a_G^- = \frac{G-1}{G}p$, giving $\Omega_G = (G-1)/G$. For `rloo`, $A_1 = r_1 - \frac{1}{G-1}\sum_{i\geq 2} r_i$, so $a_G^+ = 1-p$, $a_G^- = p$ and $\Omega_G = 1$. For `grpo` with $G = 2$, the only informative group has rewards $\{0,1\}$ and advantages $\pm\frac{1}{2}/\sqrt{\frac{1}{4}+\epsilon} \approx \pm 1$, so $a_2^+ \approx 1-p$, $a_2^- \approx p$ and $\Omega_2 \approx 1$. As $G \to \infty$, $j/G \to p$ in probability, so $a_G^+ \to (1-p)/\sqrt{p(1-p)+\epsilon}$ and $a_G^- \to p/\sqrt{p(1-p)+\epsilon}$ by bounded convergence, and $\Omega_G \to (p(1-p)+\epsilon)^{-1/2}$, the weight in the recurrence of Mroueh [2025]. If the implementation uses the Bessel-corrected standard deviation, replace $\frac{c}{G}(1-\frac{c}{G})$ in Eq. (**??**) by $\frac{c(G-c)}{G(G-1)}$. □

Derivation of Eq. (??). Take $L = 1$, K answers with one correct answer c, and logits ℓ_k, so $\pi_k \propto e^{\ell_k}$. Write $g_k = w(r_k) - \beta(\log \frac{\pi_k}{\pi_{\mathrm{ref},k}} + 1)$ and $\bar{g} = \sum_k \pi_k g_k$. The expected-gradient flow on F is $\dot{\ell}_k = \pi_k(g_k - \bar{g})$. With a zero-initialized head, the $K-1$ wrong answers start with equal logits and receive equal updates, so they stay equal, each with probability $\pi_w = (1-p)/(K-1)$. Let $\Delta = g_c - g_w$. Then $\dot{\ell}_c = p(1-p)\Delta$ and $\dot{\ell}_w = -p\,\pi_w \Delta$. Using $\pi_{\mathrm{ref},w} = (1-p_{\mathrm{ref}})/(K-1)$, $\Delta = a_G^+ + a_G^- - \beta(z - \mathrm{logit}\, p_{\mathrm{ref}})$, where $z = \mathrm{logit}\, p = \ell_c - \ell_w - \log(K-1)$. Hence $\dot{z} = \dot{\ell}_c - \dot{\ell}_w = \kappa\,\Delta$ with $\kappa = p(1-p) + p\,\pi_w > 0$, which is Eq. (**??**). A fixed point is stable when $\partial\Delta/\partial z < 0$, i.e. when $\Omega_G'(p^*)\, p^*(1-p^*) < \beta$. Mroueh's map $z_n = \mathrm{logit}\, p_{\mathrm{ref}} + \Omega(\sigma(z_{n-1}))/\beta$ has slope $s = \Omega'(p^*)\, p^*(1-p^*)/\beta$ at a fixed point and is stable when $|s| < 1$. The two conditions agree when $s > 0$; when $s < -1$, the map oscillates but the flow is still stable.

B Experimental Details

Component	Setting
Prompt features	$x \sim \mathcal{N}(0, I_{16})$; 512 train / 256 held-out test prompts
Task	$K = 4$ (E1, E3–E5), $K \in \{8, 1000\}$ (E2); $L \in \{1, \ldots, 8\}$; Gaussian W_t
Mixed-length task (E3)	$L(x)$ uniform on $\{2, \ldots, 8\}$, fixed per prompt
Multi-solution task (E4)	m of K tokens correct per position; unequal initial probabilities on correct tokens
Policy	MLP (2 hidden layers, width 128) on $(x$, one-hot $t)$; zero-init head, so π_{ref} is uniform. E2 and E3 also use a tabular softmax per prompt
Optimizer	Adam (E1, E4, E5); plain SGD for E2 and the learning-rate sweep (E3)
Batch	$B = 32$ prompts $\times$ G samples; fixed-compute runs use $B \times G = 256$
Advantage std	$\sqrt{\bar{r}(1-\bar{r}) + \epsilon}$, $\epsilon = 10^{-5}$, no Bessel correction
Seeds	3 per configuration
Compute	CPU only; each run takes seconds to minutes
LLM (E6)	Qwen2.5-0.5B-Instruct; GSM8K (MATH levels 1–5 as fallback)
LLM training	TRL GRPOTrainer + vLLM; $G \leq 16$, lr 5×10^{-6}, $\beta = 0.1$, 200-token context, temperature 0.1
LLM compute	H100 MIG slices (40 GB) on DRAC Fir; $\approx$3 GPU-days budget